

Delegation cascades: why safe instructions produce unsafe agents

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Abstract

We study what happens to a safety instruction when it is passed down a chain of delegated agents before anything is done with it. A principal enters a competitive race but does not act in it: it issues a specification to an agent, which issues one to a sub-agent, and so on for d hand-offs. Two things happen along the chain, and both are geometric in d . The specification is transmitted imperfectly, so it survives d hand-offs intact with probability $(1 - \varepsilon)^d$. Responsibility is traced back only in part, so a principal at depth d is charged the fraction ϕ^d of the harm it causes. We embed both in the reduced four-strategy AI race of Fernández Domingos and Han, make delegation depth part of the strategy, and analyse the resulting dynamics. Four results follow. First, depth is an unbounded liability shelter. Realised harm saturates while attributed harm decays, so for every $\phi < 1$ there is a depth beyond which each further hand-off strictly lowers the bill even though it strictly raises the damage. No level of liability stops chains from lengthening past it. Second, and against the intuition that the two mechanisms simply add up, neither is dangerous on its own. Start from a race that liability has already made safe at depth zero. Imperfect transmission alone lifts the long-run unsafe frequency to 0.018 and imperfect attribution alone to 0.012; together they lift it to 0.322, so 91% of the joint effect is an interaction. A counterfactual that freezes the population composition isolates its single cause. Under full pass-through the market answers transmission loss by *shortening* its chains, from six hand-offs to two, and per-layer attribution is precisely what disables that answer. Third, the resulting equilibrium is one in which every principal issues the safest available instruction and the population executes the unsafe designs anyway: declared and executed behaviour differ by 0.74 in total variation. Fourth, instruments that act on the incentive to delegate behave monotonically while instruments that act on transmission fidelity do not. A partial fidelity improvement can leave the ecosystem worse off than no improvement at all, because over-specification was an equilibrium response to erosion. A ceiling of one hand-off removes the unsafe behaviour entirely and raises social payoff from -0.8 to 60.5 .

Keywords: evolutionary game theory; delegation; principal-agent chains; liability; AI governance; multi-agent systems

1 Introduction

Software that acts on behalf of someone else is increasingly built as a chain. A person instructs an agent, the agent decomposes the task and instructs sub-agents, and the sub-agents call further

tools and agents. The process that finally takes an action in the world is then several hand-offs away from the party that wanted it taken [1, 2]. Governance of such systems is written almost entirely at the two ends of the chain. Safety training and evaluation act on the model; liability, procurement rules and disclosure requirements act on the principal. The hand-offs in between are treated as plumbing.

They are not plumbing. A hand-off does two things that matter. It transmits a specification imperfectly, because each layer restates the task in its own terms and the part most easily dropped is the qualifying clause rather than the task itself. Degradation under repeated re-encoding is a robust property of transmission chains, in human cultural transmission [3] and in recursive machine generation [4]. And it attenuates responsibility, because harm that reaches the world through several intermediaries is traced back to the party at the top only in part. Both effects are per-layer, so both compound geometrically with the number of hand-offs. What is not obvious is what a population of principals does about them.

The question matters because the two effects push the principal in opposite directions. Imperfect transmission is a private cost: a principal who is held responsible for what its chain does has every reason to keep the chain short. Imperfect attribution is a private benefit: depth buys a discount on the bill. Which force wins is not a matter of taste. Neither is the answer to the policy question that follows: whether an instrument that improves transmission, one that restores attribution, or one that simply caps the length of the chain does more for the same effort.

We answer both by making delegation depth a strategic variable in an evolutionary model. The interaction layer is the reduced four-strategy AI race of Fernández Domingos and Han [5], used without modification, so that every depth effect we report is attributable to the delegation layer we build on top of it. A design is a pair (d, s) : a number of hand-offs and the specification the principal issues. One hand-off acts on the specification by a row-stochastic kernel, and the harm a chain causes is charged to its principal at the rate ϕ^d . Principals do not optimise; designs that pay better are copied more, in the standard replicator and finite-population dynamics of evolutionary game theory [6–8].

Our contribution is an explanation, not a method or a benchmark. In one sentence: *delegation depth is a strategic variable whose two effects are individually minor and jointly decisive, because per-layer attribution disables the market’s own response to specification erosion.* The results below make that precise.

Results. We show, first, that depth is an unbounded liability shelter (Theorem 4). Realised harm saturates in d because the specification is eventually forgotten, while attributed harm carries the extra factor ϕ per layer. For every $\phi < 1$ there is therefore a depth beyond which each further hand-off strictly reduces the harm the principal is charged for while strictly increasing the harm it causes. Beyond that depth the invasion condition for one more layer stops depending on liability in the usual direction: raising the penalty makes the deeper design *more* attractive, not less (Proposition 5).

Second, the two mechanisms do not add. We calibrate the liability so that the depth-zero race is already safe: the effective liability we use is 3.1 times the level at which the single-layer game has long-run unsafe frequency zero. At that setting, imperfect transmission alone yields 0.018 and imperfect attribution alone yields 0.012, while the two together yield 0.322. The interaction, +0.293, is 91% of the joint effect. A counterfactual that holds the population composition fixed

shows why. Under full pass-through the population answers transmission loss by shortening its chains from six hand-offs to two. That answer, and not any change in behaviour at a given depth, is what per-layer attribution removes.

Third, the equilibrium that results is one in which safety declarations are uninformative. Every principal issues **AS**, the safest available specification, and the population executes **AS** only 26% of the time; the distance between what is declared and what is executed is 0.74 in total variation. Over-specification is the equilibrium response to erosion, and it does not repair it.

Fourth, the four governance instruments we compare are not interchangeable. Restoring attribution has a threshold rather than a gradient: nothing happens until ϕ reaches 0.86, after which chains collapse from six hand-offs to two. Capping chain length is monotone and cheap, and a ceiling of one hand-off both removes the unsafe behaviour and raises social payoff from -0.8 to 60.5 . Improving transmission fidelity is neither monotone nor safe in the small. Taking the erosion probability from 0.20 to 0.07 reduces the unsafe frequency to 0.032, but taking it to 0.04 raises it again to 0.239, because the improvement removes the reason principals were over-specifying. Where a single check is placed in the chain matters by a factor of 18, and it should be placed at the agent that acts.

Assumptions. Three should be stated at the outset rather than at the end. The chain transmits one specification per race, so the executed designs of the two seats are independent draws and every payoff is bilinear; a model in which layers re-negotiate during play would not have this structure. The erosion kernel is exogenous and is not itself under selection, so we describe what a market does under a given transmission technology, not how that technology evolves. And attribution decays geometrically, which is a modelling choice that we treat as the definition of per-layer liability rather than as an empirical claim; Section 6 reports what changes when it is replaced.

2 Related work

Evolutionary models of AI races. The idea of treating a development race between AI teams as an evolutionary game is due to Han et al. [9]. The reduced four-strategy race we use as our interaction layer is the model of Fernández Domingos and Han [5], in which participants repeatedly choose Safe or Unsafe, unsafe development pays more per round and advances faster, and a leader who cut corners carries a private risk of a setback. That line of work has since examined regulation, voluntary commitments and monitoring as ways of moving the equilibrium [10]. All of it places the strategic agent in the race. Our contribution is orthogonal: we leave the race exactly as it is and ask what happens when the party whose design is being selected is several hand-offs removed from the party that acts.

Delegation to artificial agents. Behavioural work has shown that delegating a decision to an artificial agent changes what people choose, sometimes towards more prosocial outcomes [11]. That literature studies a single delegation step, from a human to a machine. The governance literature has begun to describe multi-step agentic delegation and the accountability problems it raises [12–14], but as far as we are aware without a formal model in which the depth of the chain is itself selected. Making depth a strategic variable is what allows the two mechanisms to be separated and their interaction to be measured.

Hierarchies, control loss and responsibility. That authority degrades as it passes down a hierarchy is a classical observation in the economics of organisation. It appears as the loss of control that bounds firm size [15], as the cost of decentralised information processing [16], and as collusion between layers [17]. Diluted responsibility in teams is equally classical [18], and the law has long recognised that liability must sometimes be assigned to a party other than the actor for incentives to survive an intermediary [19, 20]. Our model brings the two together and adds the ingredient those literatures do not have: a population in which the depth of the hierarchy is not chosen by an optimising designer but selected by differential replication of designs.

Specification and alignment. The gap between a stated objective and an executed one is the subject of the specification-gaming literature [21], and incomplete contracting has been proposed as the right frame for alignment [22]. Those accounts locate the gap between the principal and a single agent. We locate a second gap, which compounds with the number of intermediaries and which no amount of care at either end of the chain removes.

3 Model

3.1 The interaction layer

Two seats play the repeated race of Fernández Domingos and Han [5]. In each round a seat plays Safe or Unsafe. Unsafe pays more in the stage game, with round payoffs 1.0 and 0.6 for a safe player against a safe and an unsafe opponent against 2.4 and 2.0 for an unsafe one, and it contributes 1.5 rather than 1 to race progress. The race lasts T rounds, with T guaranteed to be at least five and stopping with probability 0.2 per round thereafter, so $\mathbb{E}[T] = 9$. The leader in accumulated progress at the horizon takes a prize of 100, split on a tie, and a seat that is at risk loses everything it has accumulated with probability $p_{\max}n^U/T$, where n^U is its number of unsafe rounds and $p_{\max} = 0.6$.

Following the source study we work with four reduced designs, written here in the order in which their specifications lose clauses: **AS** is unconditionally safe, **CS** opens Safe and then copies the opponent, **CAS** opens Unsafe and then copies, and **AU** is unconditionally unsafe. All four are deterministic, so the action path of an ordered pair is fixed once the pair is fixed and the only stochastic primitive is the horizon. We therefore evaluate every matchup by exact expectation over the horizon law rather than by sampling, which removes simulation noise from the payoff matrix entirely. Write $a(i, j)$ for the expected task payoff of design i against j , $m(i, j)$ for its expected number of Unsafe actions and $u(i, j)$ for its expected fraction of Unsafe rounds.

Reading down any column of m in Table 1, the entries never fall. This monotonicity is the structural fact on which everything below rests, and we record it as a property of the interaction layer rather than an assumption of ours.

Lemma 1 (The erosion order is a harm order). *For every opponent j , $m(\text{AS}, j) \leq m(\text{CS}, j) \leq m(\text{CAS}, j) \leq m(\text{AU}, j)$, and the same holds for u . The property is verified numerically for every prize in $\{50, 100, 200\}$ and every $p_{\max}$ in $\{0.1, 0.3, 0.6, 0.9\}$.*

Table 1: Interaction layer at depth zero. Task payoff $a(i, j)$ and expected number of Unsafe actions $m(i, j)$ of the focal design i against j , evaluated exactly over the horizon law. Reading down a column, m never falls: the erosion order is a harm order.

i	$a(i, j)$				$m(i, j)$			
	AS	CS	CAS	AU	AS	CS	CAS	AU
AS	59.00	59.00	8.60	5.40	0.00	0.00	0.00	0.00
CS	59.00	59.00	26.64	16.60	0.00	0.00	4.22	8.00
CAS	101.77	61.63	27.20	27.20	1.00	4.78	9.00	9.00
AU	48.64	47.36	27.20	27.20	9.00	9.00	9.00	9.00

Table 2: The two mechanisms, separately and together. Long-run Unsafe frequency U , mean delegation depth $\bar{d}$, declaration gap and social payoff in the stationary regime of the finite-population process. Neither mechanism does much alone; the interaction is 0.293, or 91% of the joint effect.

erosion ε	attribution ϕ	U	$\bar{d}$	gap	π_S
0	1	0.0000	6.00	0.000	68.0
0.2	1	0.0176	1.99	0.358	58.3
0	0.75	0.0115	5.95	0.000	65.5
0.2	0.75	0.3225	6.00	0.738	-0.8
interaction		+0.2934			

Table 3: Depth ceilings. Long-run Unsafe frequency and social payoff when chains longer than $\bar{D}$ hand-offs are forbidden. The social payoff is maximised well inside the ceiling the market selects.

$\bar{D}$	U	$\bar{d}$	π_S
0	0.0000	0.00	59.0
1	0.0000	1.00	60.5
2	0.0178	2.00	58.3
3	0.0640	3.00	49.9
4	0.1359	4.00	36.0
5	0.2251	5.00	18.5
6	0.3225	6.00	-0.8

3.2 The delegation chain

A seat is operated not by a principal but by a chain

$$\text{principal} \longrightarrow \text{agent}_1 \longrightarrow \cdots \longrightarrow \text{agent}_d \longrightarrow \text{the race},$$

where $d \geq 0$ is the *delegation depth*. The principal issues one specification, its *intent* s , and each hand-off transmits it through the row-stochastic kernel

$$M = (1 - \varepsilon)I + \varepsilon D, \tag{1}$$

with ε the per-hand-off erosion probability and D a drift kernel on the four designs. The identity term in (1) is the part of a specification that survives a hand-off untouched. Our baseline D

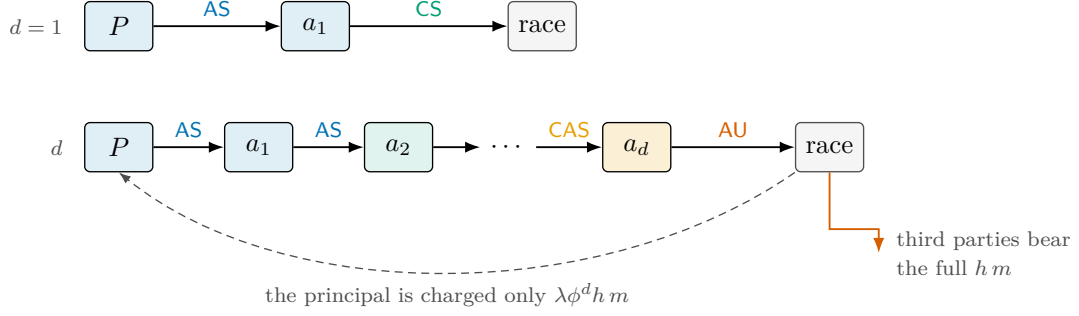


Figure 1: A delegation cascade. A principal P issues one specification and never acts. Each hand-off may drop a safety clause, so the design that reaches the race is a random function of the intent and of the depth alone; the deeper the chain, the further the executed design has slid down the erosion order $AS \rightarrow CS \rightarrow CAS \rightarrow AU$. The harm the chain causes falls on third parties in full, but is charged back to the principal at the rate ϕ^d . Both effects are per-layer, so both are geometric in the depth, and they pull the principal in opposite directions: erosion makes depth expensive, attenuated attribution makes it cheap.

is the *ladder*: an erosion event moves the specification one rung down the order AS , CS , CAS , AU , which is to say it drops one safety clause and leaves the description of the task intact. Two alternatives are used as controls. Under *collapse* an erosion event discards the specification and the sub-agent falls back on the locally competitive design AU . Under *uniform* it replaces the specification with a uniformly random one; that kernel has the same spectral gap as the ladder but no direction, and it separates the effect of biased erosion from the effect of noise as such.

The executed design of a depth- d chain with intent s is therefore distributed as row s of M^d , written $p_{d,s}$. Figure 1 shows the construction, and Figure 2 shows what it does to an AS instruction.

We also allow one *check* in the chain. A check of strength α at layer $k \leq d$ restores the intent with probability α , after which the specification still has $d - k$ hand-offs to survive, so the executed design is distributed as $\alpha M^{d-k}(s, \cdot) + (1 - \alpha)M^d(s, \cdot)$.

3.3 Three functionals

Two designs meeting in the race produce, in expectation over the executed designs and over the horizon,

$$a(d, s; d', s') = p_{d,s}^\top A p_{d',s'} + B(d), \quad m(d, s; d', s') = p_{d,s}^\top M_h p_{d',s'}, \quad (2)$$

where A and M_h are the task-payoff and unsafe-action matrices of the interaction layer and $B(d) = g d - c d^2$ is the net organisational benefit of running a chain of depth d . Everything is bilinear because one specification is transmitted per race and then played, so the executed designs of the two seats are independent draws. The benefit B is real: delegation genuinely produces more, and the term enters the social accounting as well as the private one. The baseline sets $g = 1.5$, about 2.5% of the payoff of a safe seat per hand-off, and $c = 0$, so that in the baseline the only force limiting depth is the harm the principal internalises.

Let h be the external harm of one Unsafe action and λ the liability rate at depth zero. The *private* functional, which alone drives the dynamics, and the *social* functional, which alone scores

them, are

$$\pi_P(d, s; d', s') = a(d, s; d', s') - \lambda \phi^d h m(d, s; d', s'), \quad (3)$$

$$\pi_S(d, s; d', s') = a(d, s; d', s') - h m(d, s; d', s'), \quad (4)$$

The two functionals in (4) differ only in the second term, and their difference is the *dilution wedge*

$$\Delta(d) = \pi_P - \pi_S = h(1 - \lambda \phi^d) m, \quad (5)$$

The wedge (5) is non-decreasing in d for every $\phi \leq 1$ and vanishes identically at $\phi = \lambda = 1$. Depth alone widens the gap between what a principal optimises and what society bears, without any change in what the principal intends. Only the product $L = \lambda h$ enters π_P , so the two-parameter family (λ, h) collapses to the single control L , the *effective liability*; h is a property of the world and stays fixed when the instrument moves, which is what keeps social payoffs comparable across liability regimes.

3.4 Dynamics and the choice of parameters

Designs are copied in proportion to how well they pay. We report the stationary regime of the finite-population pairwise-comparison process in the small-mutation limit [23, 24] as the primary reading, with population size $Z = 100$ and selection intensity $\beta = 0.05$, and the basin-averaged attractor of the replicator equation as a secondary one. The joint design space has $4(D + 1)$ elements, 28 at the baseline ceiling $D = 6$, which puts the full mutation-selection chain out of reach: its state space has $\binom{Z+n-1}{n-1}$ elements. We therefore compute fixation probabilities in closed form and check the reduction against the full chain on a reduced design space (Section 4.7), where the two agree to 0.012 at the smallest mutation rate examined.

The remaining choice is the liability level, and we fix it so that the delegation results cannot be an artefact of a race that was unsafe to begin with. Bisection on the depth-zero game puts the long-run unsafe frequency below 10^{-3} for every $L \geq 6.52$; we run at $L = 20$, that is 3.1 times the level at which the single-layer race is already safe. At that level the depth-zero ecosystem has long-run unsafe frequency 0.000 under the finite-population reading and 0.026 under the replicator reading. The baseline erosion probability is $\varepsilon = 0.20$ and the baseline attribution retention is $\phi = 0.75$; both are swept over their full ranges in Section 4.4 and Section 4.7.

4 Results

4.1 The specification is forgotten geometrically

Under the ladder kernel the number of clauses lost over d hand-offs is binomial, so the executed design is the intent shifted down the erosion order by $\text{Bin}(d, \varepsilon)$ rungs and truncated at AU. Two consequences are exact.

Proposition 2 (Erosion law). *For any intent other than the absorbing one, the specification survives d hand-offs intact with probability exactly $(1 - \varepsilon)^d$, and the influence of the intent on the executed design decays at the rate given by the second-largest eigenvalue modulus of M , which for the ladder and collapse kernels equals $1 - \varepsilon$. The specification half-life $d_{1/2} = \ln 2 / \ln(1/(1 - \varepsilon))$ therefore depends on nothing but the per-hand-off erosion probability.*

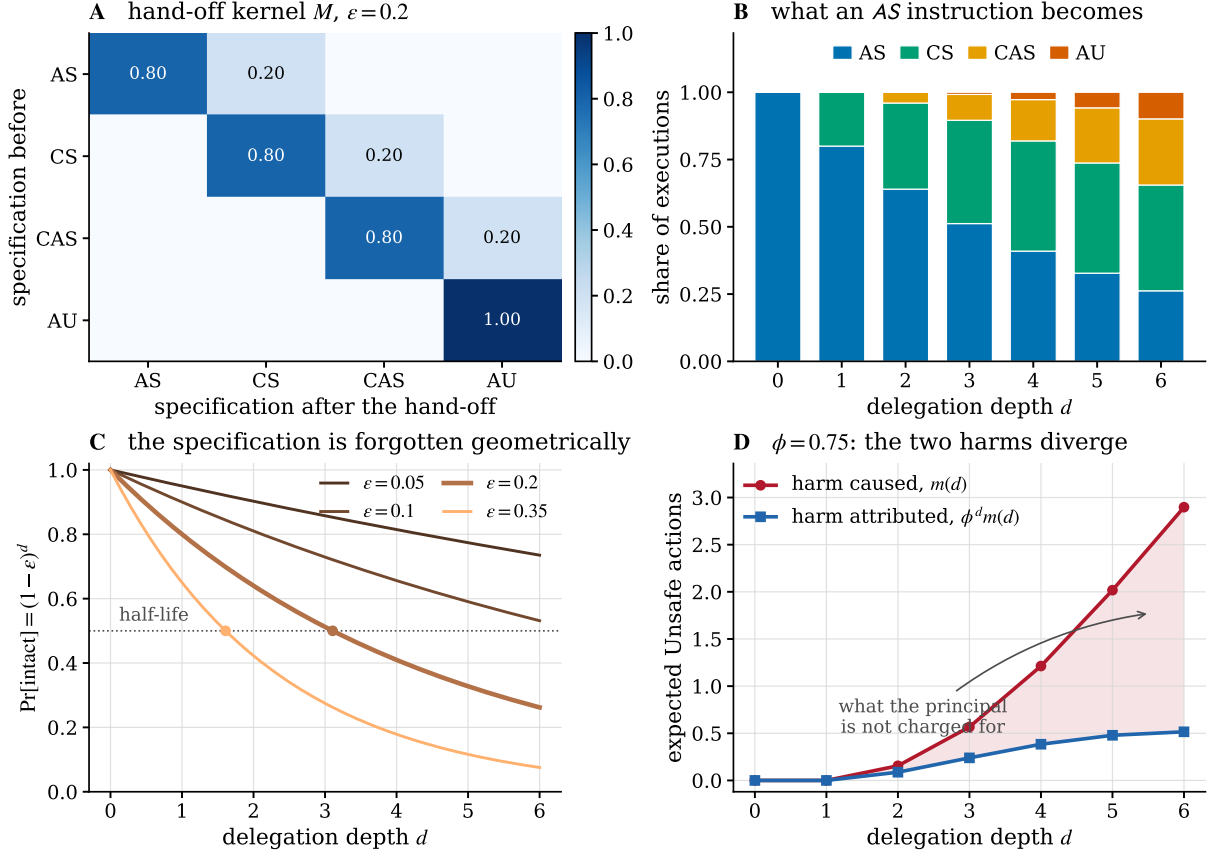


Figure 2: What a hand-off does to a specification. (A) The hand-off kernel at the baseline erosion probability: one clause is lost per erosion event, and the unconditionally unsafe design is absorbing. (B) The executed design of a chain that was told AS, by depth. At six hand-offs the instruction is executed as issued in 26% of cases and has slid to CAS or AU in 34%. (C) The probability that a specification arrives intact is exactly $(1 - \varepsilon)^d$; dots mark the half-life, 3.11 hand-offs at $\varepsilon = 0.20$. (D) Harm caused and harm attributed for the same chain. The shaded wedge is the part of the damage that never reaches the principal’s accounts.

At the baseline $\varepsilon = 0.20$ the half-life is 3.11 hand-offs and only 26.2% of specifications reach a six-deep executor unaltered (Figure 2B,C). The claim is not that instructions are forgotten quickly; it is that they are forgotten at a rate that no amount of care at the top of the chain changes, because the decay is in the number of hand-offs and not in the effort spent writing the instruction.

Because the ladder kernel only moves mass down the erosion order, and because by Lemma 1 harm is monotone along that order, the harm a chain causes is monotone in its depth.

Proposition 3 (Depth monotonicity). *Let M be a hand-off kernel that is stochastically monotone and satisfies $\delta_s \preceq M(s, \cdot)$ for every s in the erosion order. Then for every intent s and every opponent mixture, $d \mapsto m(d, s; \cdot)$ is non-decreasing. In particular the ladder and collapse kernels satisfy the hypothesis; the uniform kernel does not.*

Proof sketch. $\delta_s \preceq M(s, \cdot)$ and stochastic monotonicity of M give $M^d(s, \cdot) \preceq M^{d+1}(s, \cdot)$ in the usual stochastic order by induction. Lemma 1 makes $m(\cdot, j)$ a non-decreasing function on the order for every j , and the expectation of a non-decreasing function respects the stochastic order. Bilinearity in (2) extends the statement from pure opponents to mixtures. $\square$

4.2 Depth is a liability shelter

The harm a chain causes is bounded: once the specification has been forgotten, the executed design is the absorbing one whatever the intent, so $m(d)$ converges. The harm a chain is charged for carries the extra factor ϕ per layer and is not bounded below by anything. Comparing the two gives the central structural result of the model.

Theorem 4 (Liability shelter). *Adding a hand-off lowers the attributed harm exactly when*

$$\phi < \phi^*(d) = \frac{m(d)}{m(d+1)}. \quad (6)$$

Because m is non-decreasing and bounded, $\phi^(d) \rightarrow 1$. Hence for every $\phi < 1$ there exists a depth $\bar{d}(\phi)$ such that every hand-off beyond $\bar{d}$ strictly reduces the harm attributed to the principal while strictly increasing the harm it causes, and*

$$\lim_{d \rightarrow \infty} \phi^d m(d) = 0, \quad \lim_{d \rightarrow \infty} m(d) = m_{\max}.$$

Probing (6) to depth 40 puts ϕ^* at 0.99993 and, at the baseline $\phi = 0.75$, the shelter opening at $\bar{d} = 6$ (Figure 3A,B). The theorem is the reason the model has no interior solution to appeal to: a principal who can add layers can drive its attributed harm to zero while driving its real harm to the maximum, and the only thing that stops it is a rule that limits the number of layers or an attribution rule with $\phi = 1$ exactly.

The consequence for the liability instrument is sharp. In a resident population of depth- d principals with intent s , the advantage of a depth- $(d+1)$ mutant is affine in the effective liability,

$$\pi_P(d+1, d) - \pi_P(d, d) = \underbrace{a(d+1, d) - a(d, d)}_{\text{benefit}} - L \underbrace{[\phi^{d+1} m(d+1, d) - \phi^d m(d, d)]}_{\text{attributed gain}}, \quad (7)$$

so the single ratio implied by (7) decides the outcome.

Proposition 5 (Invasion by depth). *The critical effective liability at which depth $d+1$ loses its advantage over depth d is $L^*(d) = \text{benefit}/\text{attributed gain}$. When the attributed gain is positive, $L > L^*(d)$ blocks the deeper design. When the attributed gain is negative – which by Theorem 4 happens for all d large enough whenever $\phi < 1$ – the advantage is increasing in L , and no level of liability blocks the deeper design.*

Figure 3C shows the attributed gain for each intent at the baseline. For the intent CAS it is negative at every depth, so for those principals liability is not a tax on delegation but a subsidy for it.

Corollary 6 (When a rule calibrated at depth zero stops binding). *A liability L that clears a threshold L_c for principals who act for themselves stops clearing it at depth $d^\dagger = \log(L_c/L)/\log \phi$, which is finite for every $\phi < 1$. At the baseline, $L_c = 6.52$, $L = 20$ and $\phi = 0.75$ give $d^\dagger = 3.9$: the rule survives three hand-offs and fails at the fourth.*

The last panel of Figure 3 records a fact that is easy to miss. The depth the population reaches is not the depth that is best for the principals themselves. A monomorphic population of one-hand-off principals earns more than a monomorphic population of six-hand-off principals, and yet the deeper design invades at every step, because an invader is judged against the resident environment rather than against its own. Delegation depth is a social dilemma at the private level before it is one at the social level.

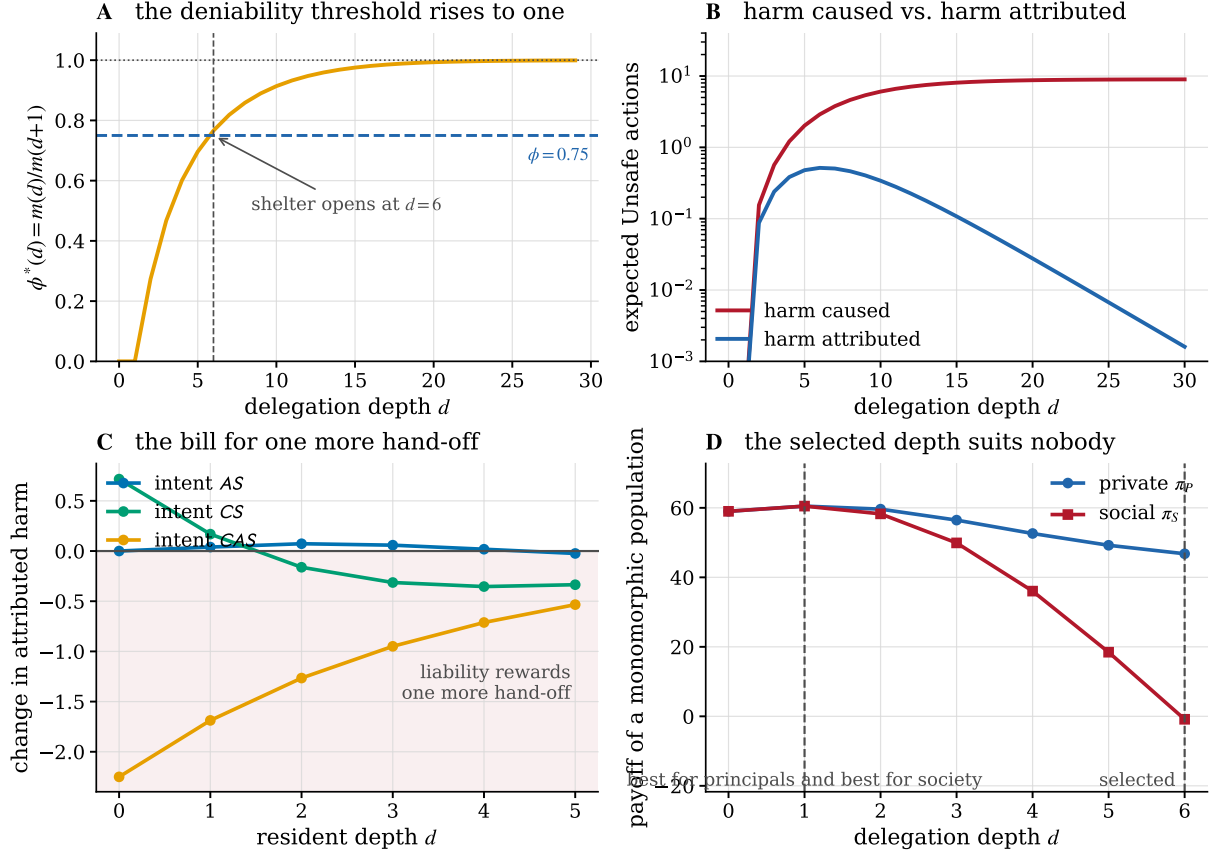


Figure 3: Depth as a liability shelter. (A) The deniability threshold $\phi^*(d) = m(d)/m(d+1)$ rises towards one because realised harm saturates, so any attribution retention below one is eventually below the threshold; at $\phi = 0.75$ the shelter opens at six hand-offs. (B) Beyond that depth, realised harm still climbs towards its ceiling while attributed harm falls geometrically. (C) What one more hand-off adds to the principal's bill. Where the curve is negative, liability rewards delegation instead of taxing it. (D) Private and social payoff of a monomorphic population by depth. Selection ends at the ceiling, which is worse for the principals than one hand-off and much worse for everybody else.

4.3 Neither mechanism is dangerous alone

The two mechanisms can be switched off independently: $\varepsilon = 0$ removes erosion and $\phi = 1$ removes attenuated attribution. Crossing them gives a 2×2 design whose four cells are reported in Table 2 and Figure 4.

Starting from a race that is already safe, erosion alone raises the long-run unsafe frequency to 0.018 and attenuated attribution alone to 0.012. An additive reading predicts 0.030 when both are present. The measured value is 0.322. The interaction, +0.293, is 91% of the joint effect, and the same picture appears under the replicator reading, where the interaction is +0.331.

The mechanism behind the interaction is visible in the second panel of Figure 4, and it is not what one would guess. Under full pass-through, erosion is a private cost, and the population responds to it by *shortening* its chains: mean depth falls from 6.00 to 1.99. Erosion is self-limiting when the principal is charged for it. Attenuated attribution does not make behaviour worse at a given depth; it removes the response. With $\phi = 0.75$ the chains stay at the ceiling and the same erosion now acts over six hand-offs instead of two.

A counterfactual makes the decomposition exact. Scoring the pass-through population with

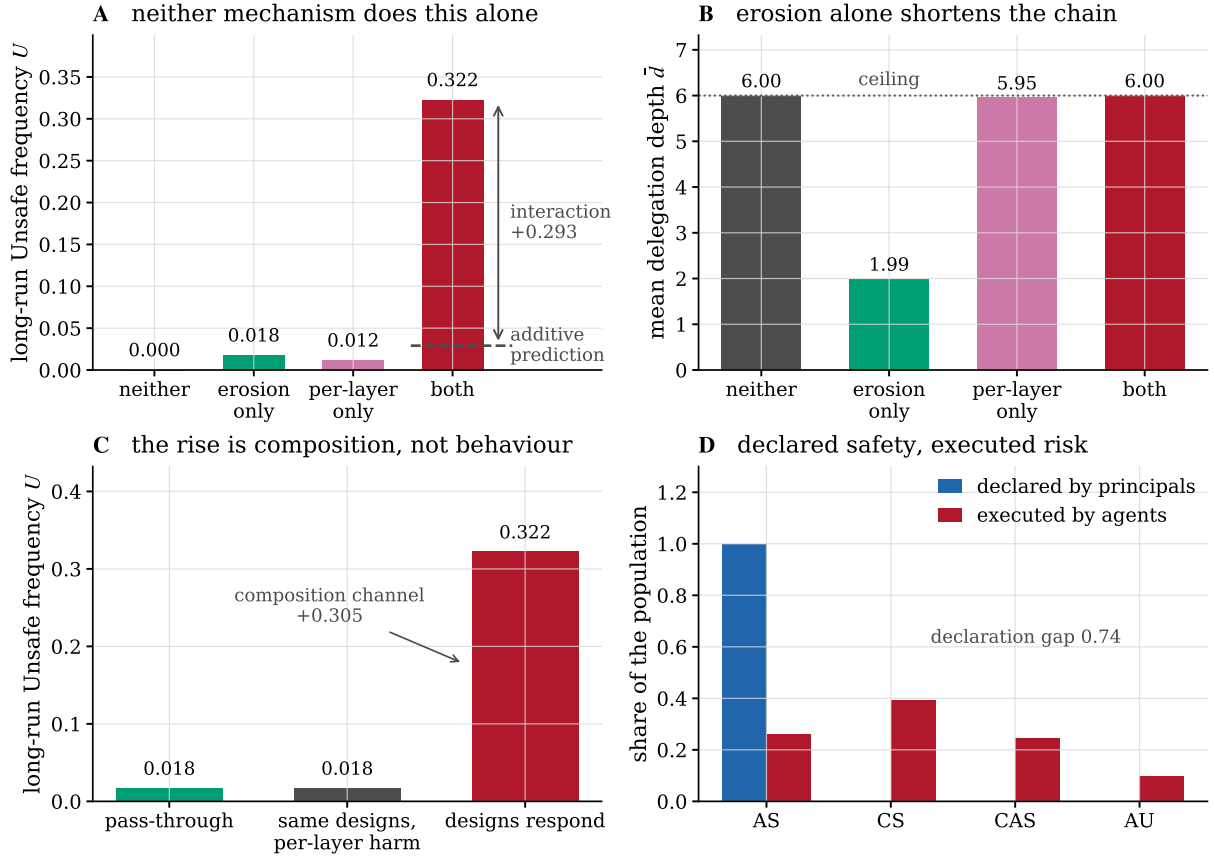


Figure 4: The two mechanisms, separately and together. (A) Long-run unsafe frequency in the four cells of the 2×2 . The additive prediction from the two single-mechanism cells is 0.030; the joint outcome is 0.322. (B) Mean delegation depth in the same four cells. Under pass-through the population answers erosion by shortening its chains; per-layer attribution removes that answer and the ceiling binds. (C) Holding the pass-through population fixed and charging it the per-layer bill changes nothing, so the entire rise is composition rather than behaviour. (D) In the joint cell every principal declares the safest instruction and the population executes something else.

the per-layer harm matrix – the same designs, charged differently – leaves the unsafe frequency at 0.018, unchanged to nine decimal places, because the behaviour of a given design does not depend on who pays for it. The whole of the remaining +0.305 is composition: which designs the population comes to contain. The result is a caution about a natural inference. Observing that deeper chains behave worse does not license the conclusion that the harm comes from erosion, because in this model almost all of it comes from the depth the market selects once erosion is no longer its problem.

4.4 The attribution–erosion plane

Neither baseline value is doing the work. Figure 5 sweeps the plane (ε, ϕ) on a 21×21 grid. The interaction index is positive over 89% of the interior and reaches 0.81; the long-run unsafe frequency reaches 0.995 and the declaration gap 0.95.

The second panel is the one to read for policy. Mean delegation depth splits the plane along a frontier: below it the ceiling binds and chains are as long as the technology allows, above it chains are short. The frontier runs diagonally, so erosion and attribution are substitutes in producing long chains – a regime with faithful hand-offs tolerates much weaker attribution before

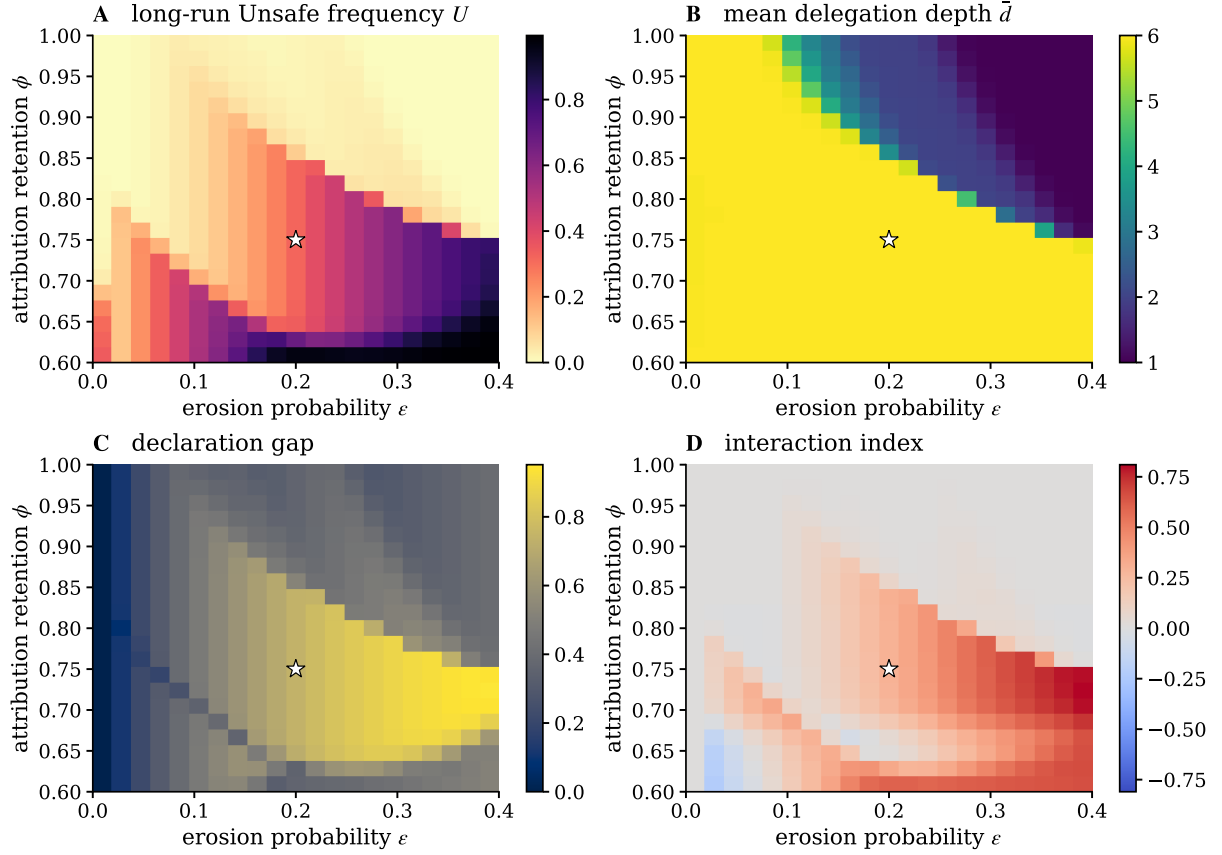


Figure 5: The attribution–erosion plane. Long-run unsafe frequency (A), mean delegation depth (B), declaration gap (C) and interaction index (D) over a 21×21 grid; the star marks the baseline. Chains are at the ceiling throughout the yellow region of (B), and the frontier that bounds it is diagonal: attribution and transmission fidelity substitute for each other in keeping chains short.

the ceiling binds, and a regime with strong attribution tolerates much noisier hand-offs.

4.5 Declared safety and executed safety

At the baseline the population is monomorphic on the design AS@6: every principal issues the safest available instruction and every principal runs a six-hand-off chain. What is executed is another matter. The realised mixture is 26% AS, 39% CS, 25% CAS and 10% AU, and the total variation distance between what is declared and what is executed is 0.74. Social payoff falls from 68.0 in the baseline cell to -0.8 .

The equilibrium is easy to misread as bad faith, and it is not. Every principal is issuing the strongest safety instruction available to it, and doing so is a best response: over-specification is how a principal compensates for a chain that will erode whatever it is told. What the equilibrium shows is that a declaration made at the top of a long chain carries almost no information about the behaviour at the bottom of it. Any governance instrument that reads declarations – self-attestation, model cards, published policies – is reading a quantity that the model predicts will be maximally safe regardless of the ecosystem it describes.

4.6 What the instruments do

Four instruments act at different points of the chain: raise the attribution retention ϕ towards pass-through; cap the permitted chain length at $\bar{D}$; make every hand-off more faithful, multiplying ε by a factor ρ ; or insert one check of strength α at a chosen layer. Figure 6 applies each to the same baseline.

Attribution is a threshold, not a dial. Raising ϕ from 0.75 does nothing at all until $\phi = 0.86$, where mean depth falls from 5.92 to 3.45 and the unsafe frequency from 0.314 to 0.103; by $\phi = 0.99$ the outcome is the pass-through outcome. The location of the jump is close to the closed form implied by Corollary 6, $\phi_c = (L_c/L)^{1/D} = 0.830$, the residual gap being the extra harm that erosion contributes at depth. Partial pass-through buys very little: at $\phi = 0.85$, which returns 85% of responsibility at each of six layers and therefore 38% overall, the ecosystem is still at 0.314.

A ceiling buys safety and welfare at once. The depth ceiling is the only instrument here that is monotone in its own strength and improves both objectives together. A ceiling of one hand-off removes the unsafe behaviour entirely and lifts social payoff from -0.8 to 60.5; a ceiling of two leaves 0.018 and 58.3. The reason it works is that the ceiling is the constraint the shelter theorem leaves standing when attribution cannot be made exact.

Fidelity is not monotone. Attestation improves matters over most of its range, from 0.322 at $\rho = 1$ down to 0.032 at $\rho = 0.35$, and then reverses: at $\rho = 0.20$ the unsafe frequency is back up to 0.239. The cause is the intent margin. Over-specification was an equilibrium response to erosion; once hand-offs are faithful enough, issuing CS rather than AS becomes the better reply, and CS eroding into CAS is far more damaging than AS eroding into CS. A partial fidelity improvement can therefore leave the ecosystem worse off than no improvement at all. The same reversal appears in the audit-placement sweep at full strength, and for the same reason; at half strength, where the intent margin does not flip, the sweep is monotone.

Check late, not early. For a fixed design the ordering is unambiguous.

Proposition 7 (Audit placement). *Under a kernel satisfying the hypothesis of Proposition 3, the realised harm of a depth- d chain carrying one check of strength α at layer k is non-increasing in k .*

Proof. The executed law is $\alpha M^{d-k}(s, \cdot) + (1 - \alpha)M^d(s, \cdot)$, and by Proposition 3 the expected harm under M^j is non-decreasing in j . Increasing k decreases $d - k$, hence decreases the first term and leaves the second unchanged. $\square$

In the equilibrium the effect is large: the same check is worth a factor of 18 more at the agent that acts than at the principal, 0.018 against 0.322. Checking the specification a party *received* tells you little about what will be done with it several hand-offs later.

Which instrument is cheapest. Asking each instrument for the same target, a long-run unsafe frequency of at most 0.05, and scoring it by the social payoff it delivers there (Figure 7B): attestation reaches the target at 58.7, the depth ceiling at 58.3, a punitive liability at 56.2 and

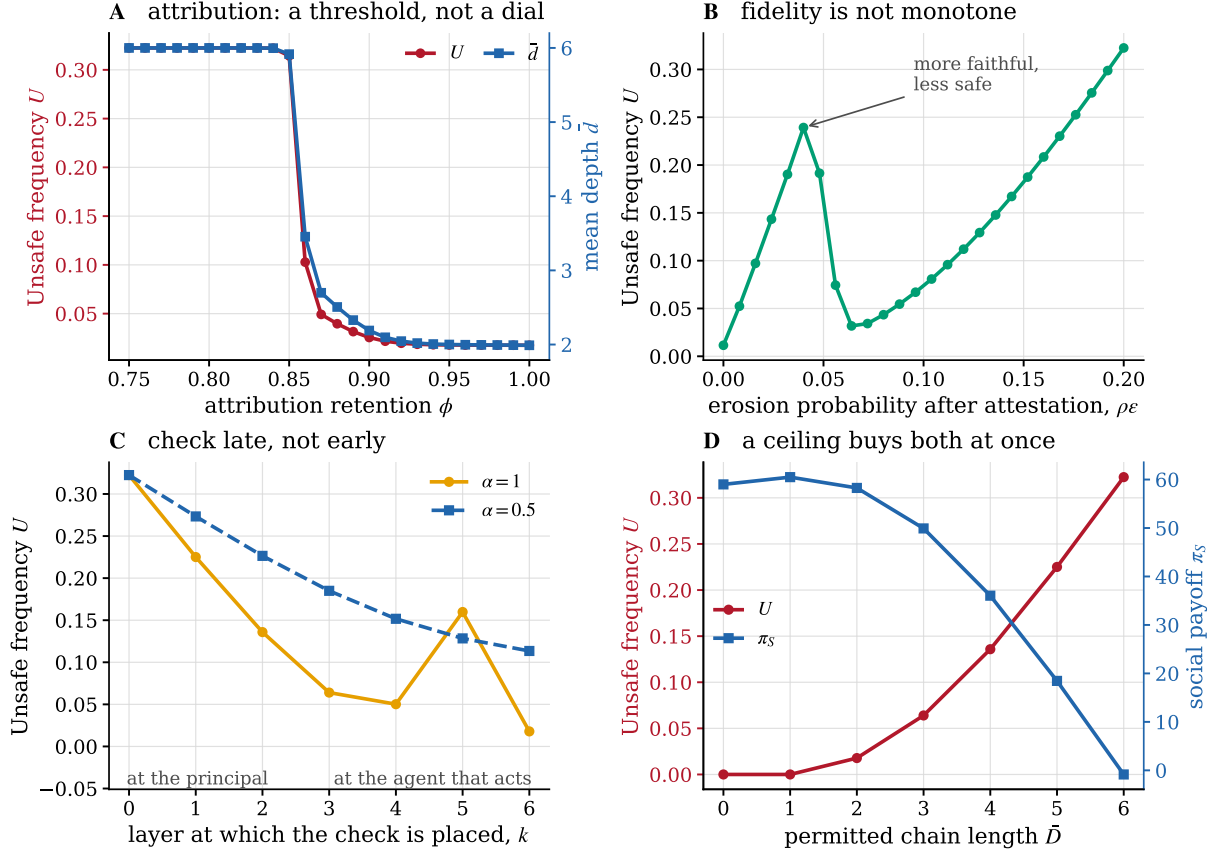


Figure 6: Four instruments. (A) Restoring attribution has a threshold at $\phi = 0.86$ and does almost nothing below it. (B) Making every hand-off more faithful is not monotone: the marked point is safer than settings with less erosion. (C) One check, moved down the chain. Late placement dominates; the non-monotonicity at full strength is the intent margin flipping, and it is absent at half strength. (D) A ceiling on chain length improves safety and welfare together.

pass-through at 53.5. The ranking is close enough that we would not press it, but the qualitative message is not: raising the liability level is the least efficient of the four, because it acts on a margin – how much harm costs – that depth has already discounted.

4.7 Robustness

The 2×2 was recomputed across 43 parameter cells: the prize and the private setback risk of the race, both readings of what a setback destroys, the organisational benefit and the coordination cost of a taller chain, the population size, the selection intensity, the depth ceiling, the liability level, and the choice between the finite-population and replicator readings. The interaction is positive in 88% of them, with median +0.292 and interquartile range [0.242, 0.297]; the median interaction share is 0.91. The mean delegation depth is higher under per-layer attribution than under pass-through in every single cell. The negative cells are all in the prize-and-risk sweep and are ceiling effects, where the joint outcome is already at or near an unsafe frequency of one and cannot show a further increase.

The drift kernel is the one dimension along which the result genuinely changes, and the direction is instructive (Figure 8B). Replacing graded erosion by *collapse*, in which a lost specification is replaced by the unconditionally unsafe design rather than by the next rung

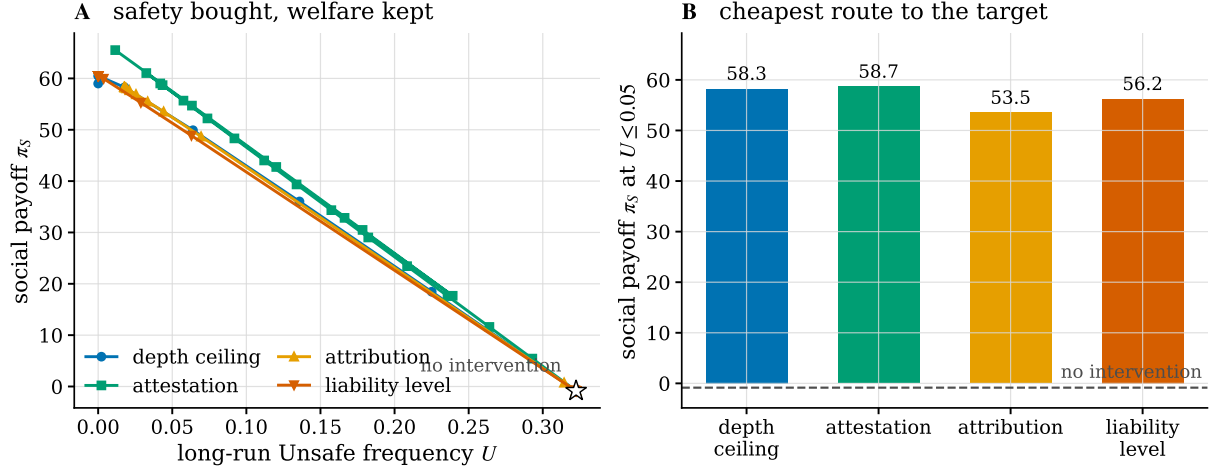


Figure 7: What each instrument costs. (A) The frontier each instrument traces in the plane of social payoff against long-run unsafe frequency, starting from the untreated baseline (star). (B) Social payoff at the weakest setting of each instrument that reaches an unsafe frequency of 0.05. Raising the liability level is the least efficient route, because depth has already discounted the margin it acts on.

down, all but eliminates the effect: the joint cell falls to 3×10^{-5} and mean depth to 0.0002. Delegation simply does not happen, because a chain that fails catastrophically is one whose principal will not build it. Replacing graded erosion by *uniform* noise, which has the same spectral gap but no direction, makes matters worse rather than better, with a joint cell of 0.517. The dangerous property of a hand-off is therefore not that it is biased but that it degrades gracefully. A specification that fails loudly deters the delegation that would have caused the harm; one that fails quietly does not.

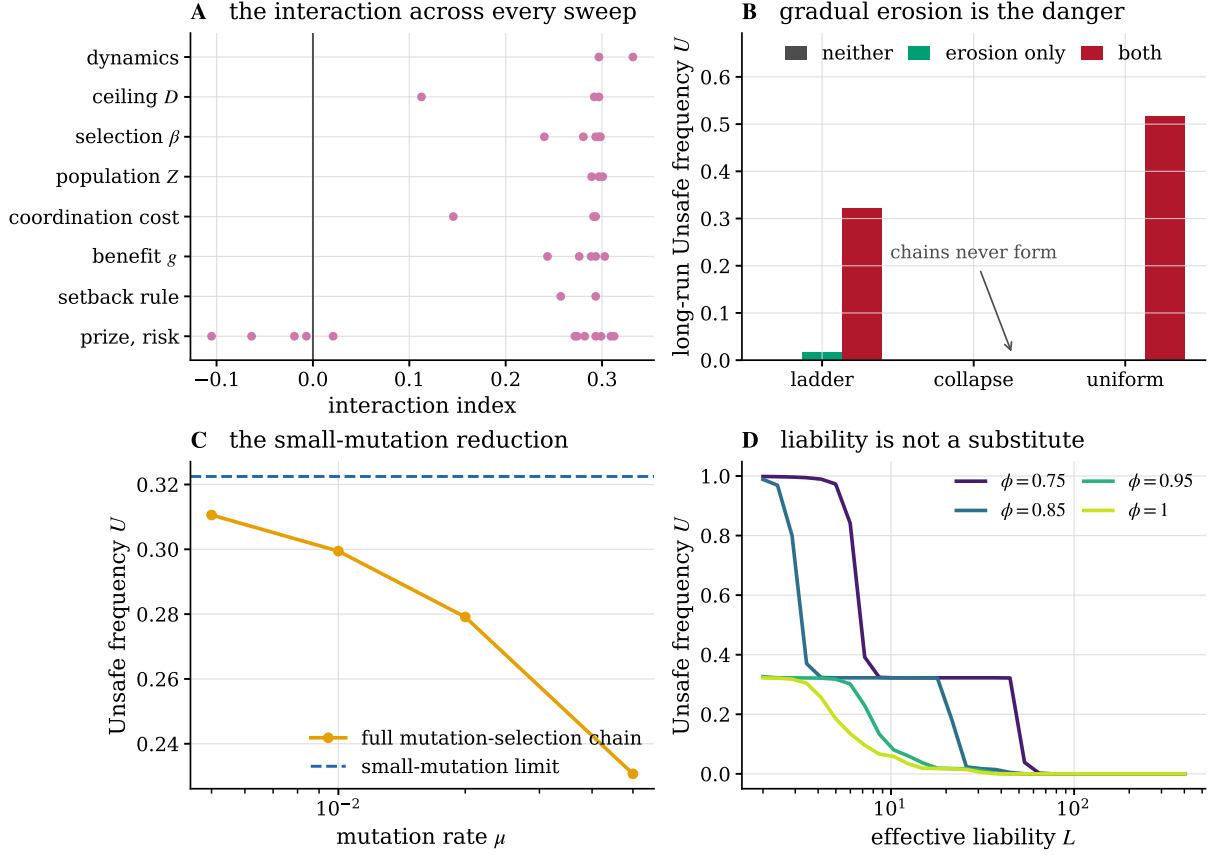


Figure 8: Robustness. (A) The interaction index across every sweep; each dot is one parameter cell. (B) Drift kernels. Graceful erosion produces the cascade; catastrophic erosion deters delegation altogether; undirected noise is worse than graded erosion, so the effect is not about the direction of the slippage. (C) The small-mutation limit against the full mutation-selection chain on a reduced design space. (D) Long-run unsafe frequency against the effective liability at four attribution retentions; a lower ϕ shifts the whole curve rather than tilting it.

5 Discussion

The model says something specific about where the governance of agentic systems is currently pointed. Almost all of the instruments now being built read or act at the two ends of a chain: evaluations and safety training at the model, and liability, disclosure and procurement rules at the deploying principal. Both ends are exactly where the model says the leverage is weakest.

At the principal end, Theorem 4 says that a liability rule written in terms of what a principal is responsible for has a depth beyond which it stops binding, and Corollary 6 says where that depth is. Raising the penalty does not move the failure depth much, because it enters logarithmically; restoring attribution does, because it enters through $\log \phi$. The practical form of that observation is unwelcome for a regulator: partial pass-through is worth very little. Recovering 85% of responsibility at each of six hand-offs recovers 38% overall and, in our baseline, changes nothing at all.

At the model end, the declaration result says that a safety commitment issued at the top of a long chain is close to uninformative about the ecosystem below it, and not because anyone is lying. In the baseline equilibrium the declaration is as strong as the language permits and the behaviour is 74% different from it. An evaluation regime that scores what a principal says,

or scores a model in isolation, is measuring a quantity that the model predicts will look good precisely when the chain is long enough for it not to matter.

That leaves two instruments with real leverage, and they are the two least discussed. The first is a limit on delegation depth. It is the constraint the shelter theorem leaves standing, it is monotone in its own strength, and in our baseline it improves welfare and safety simultaneously, which none of the others do. The second is verification placed at the point of action rather than at the point of instruction: a check of fixed strength is worth eighteen times more at the agent that acts than at the principal that spoke.

The non-monotonicity result deserves its own caution. We find that making hand-offs more faithful can raise realised harm, because the over-specification that erosion had forced becomes unnecessary and the population relaxes its declared intent by one clause, and a clause dropped from CS costs far more than a clause dropped from AS. The general form of the caution is familiar from other equilibrium settings: a partial fix to a friction that agents had been compensating for can be worse than no fix, and the sign of the effect cannot be read off the mechanism. In this model, only the instruments acting on the incentive to delegate are monotone.

Finally, the kernel comparison suggests a design principle that does not come from the liability literature at all. A specification that degrades gracefully is more dangerous than one that fails loudly, because graceful degradation lets a principal build a chain whose failures it never sees. Systems whose specifications fail closed, refusing rather than approximating when a clause is lost, restore the private cost of depth that per-layer attribution removes.

6 Limitations

The interaction layer is a stylised two-player race with four reduced strategies, inherited deliberately and without modification from Fernández Domingos and Han [5]. It is not a model of any deployed system, and the absolute numbers should be read as the behaviour of that game rather than of a market.

The erosion kernel is exogenous. We describe how a population behaves under a given transmission technology and not how the technology itself is selected; a model in which providers compete on fidelity would let ε co-evolve with d , and the non-monotonicity in Section 4.6 suggests that the outcome would not be a simple race to the top. Our chains are also linear, whereas real agentic systems branch, and a branching chain aggregates several eroded specifications rather than one.

Geometric attribution is a modelling choice. It is the natural formalisation of per-layer liability, in which each intermediary absorbs a fixed share, but real attribution regimes are lumpier: a single identifiable intermediary may absorb everything, or a strict pass-through rule may hold at some depths and not at others. What the model requires is only that attribution be non-increasing in depth and that it fall strictly at least sometimes; Theorem 4 needs $\phi^d m(d) \rightarrow 0$, which any geometrically-bounded attribution rule delivers. A rule that floors attribution at a positive constant $\phi_{\min} > 0$ would cap the shelter, and locating the floor at which the shelter closes is a natural next step.

Finally, the harm scale h enters the social functional and therefore the welfare comparisons, and we do not know it. We have kept the primary outcome, the long-run unsafe frequency, free of welfare weights for that reason, and we report social payoffs only for comparisons at a fixed h .

7 Conclusion

Delegation depth is not neutral plumbing between a principal and an action. It erodes the specification geometrically and it dilutes the attributed harm geometrically, and because the second effect is unbounded while the first saturates, depth is a liability shelter that no level of liability closes. What makes the combination dangerous is not either mechanism but the loss of a response: a principal who is charged for what its chain does answers unreliable hand-offs by keeping the chain short, and per-layer attribution is precisely what stops it. In our baseline that lost response accounts for 91% of the damage, and it produces an equilibrium in which every principal issues the safest instruction available and the ecosystem executes something 74% different.

The instruments that follow are not the ones currently receiving attention. Limiting how many hand-offs may separate an instruction from an action, and verifying specifications at the agent that acts rather than at the party that spoke, are the two levers that behave predictably in this model. Restoring attribution works, but only when it is nearly complete; raising penalties works least of all, because depth has already discounted the margin they act on.

Data and code availability

All results are deterministic given the seed used for the replicator start measure. Code, generated tables and the figure pipeline are available at github.com/<user>/delegation-cascade.

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